

DEVELOPMENT OF 2D GAME "CHESS" USING SFML

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OBJECTIVE

**Develop 2D game “Chess”
with C++ using SFML**

TASKS

- choose development tools
- game logic realization
- game testing

DEVELOPMENT TOOLS

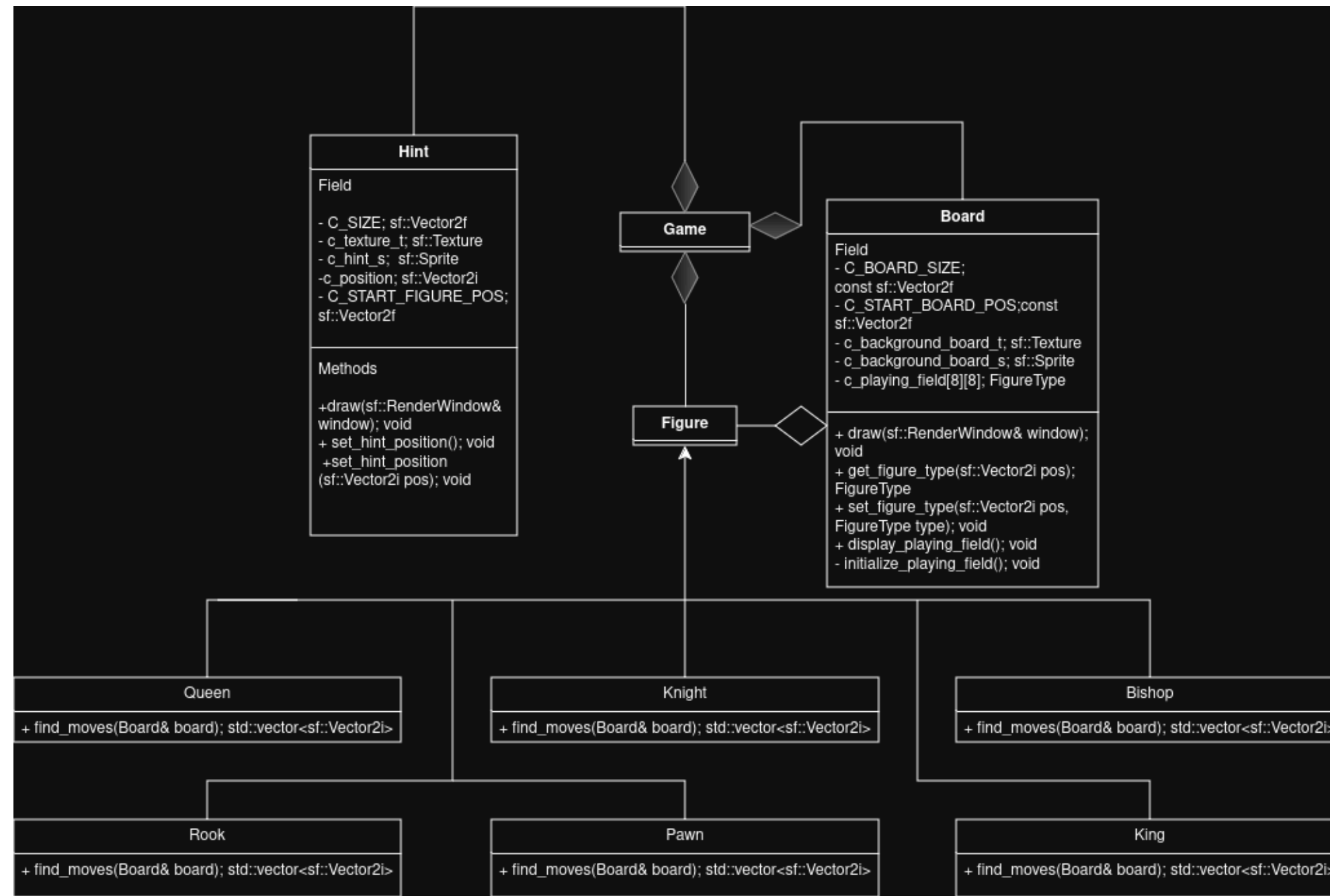
- programming language C++
- library SFML(Simple and Fast Multimedia Library)
- Visual Studio code
- LibreSprite
- GCC compiler

PROGRAM ARCHITECTURE

The program contains the following main classes:

- Board - displays playing board, simulates game process
- Figure - base class for each figure
- Game - responsible for game logic
- Hint - displays the right moves

UML diagram for game “Chess”



pic.1. UML contains all classes in game

GAME TESTING

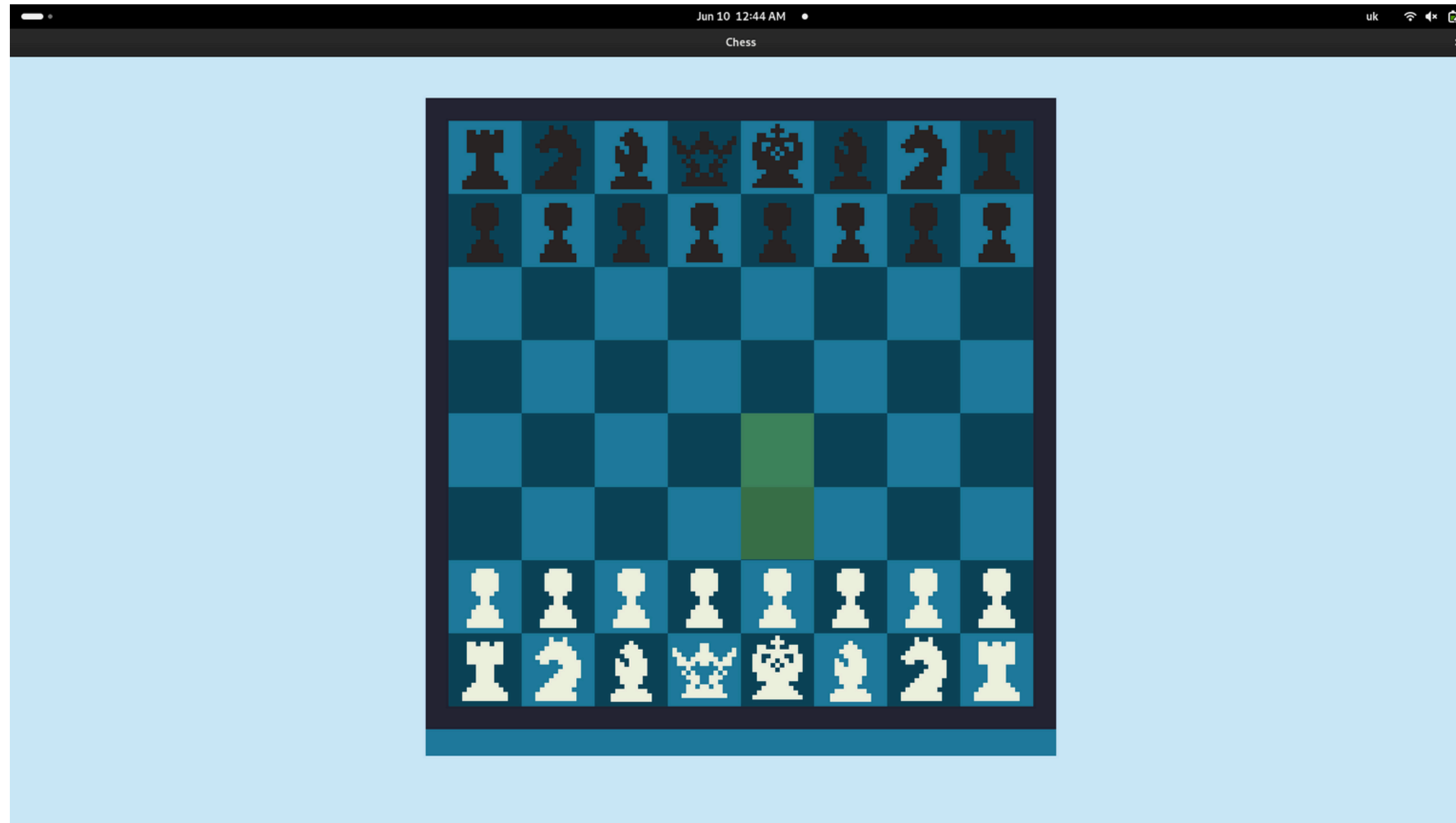
The game contains a lot of objects and game elements, which need to render. Also, during interaction, some game objects could be deleted from game process.

Object Board renders while program is running, and deletes when program close.

Other objects are dynamically created and removed during program execution.

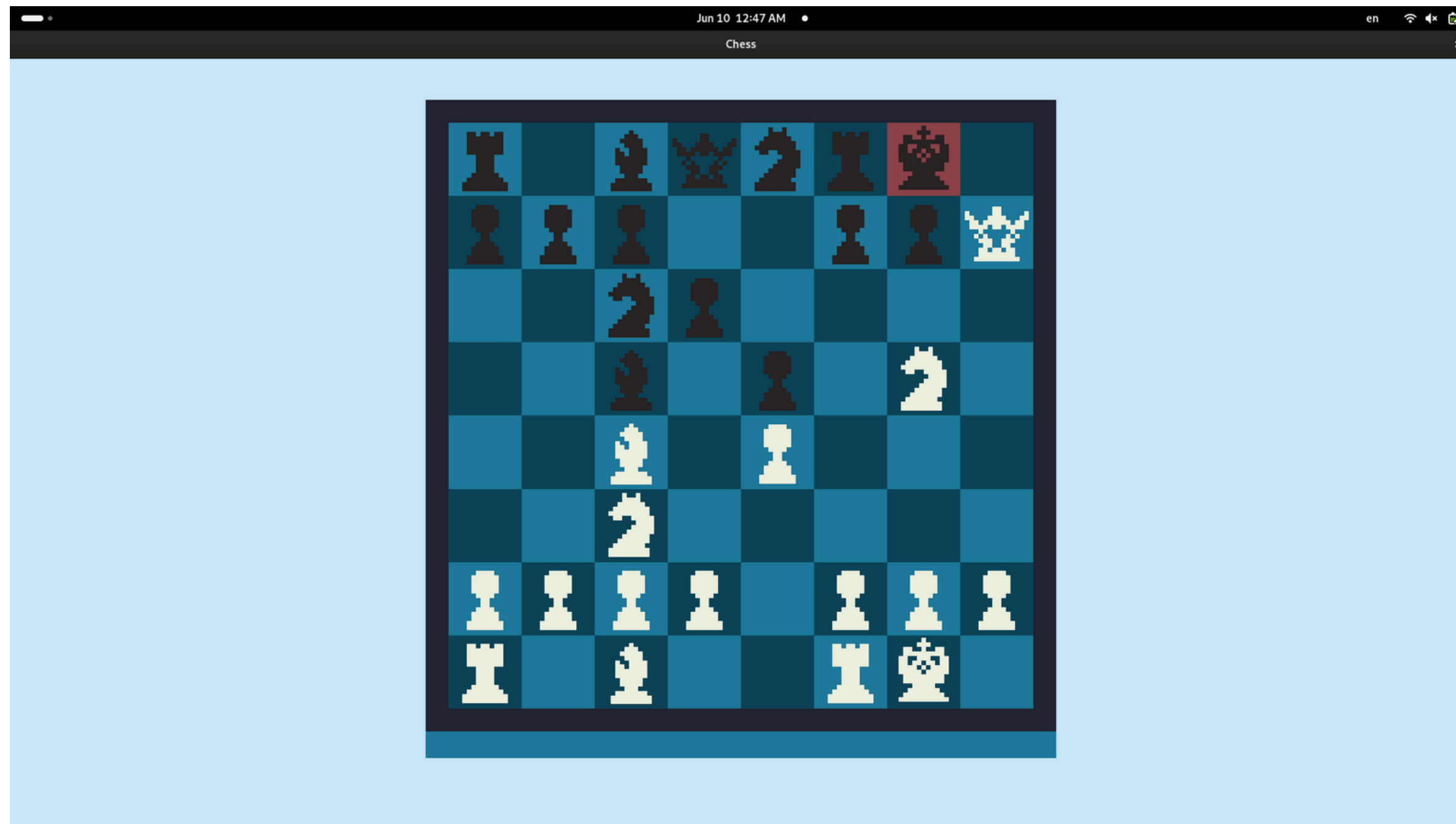


pic.2. Figure selected



pic.3.Hints displaying





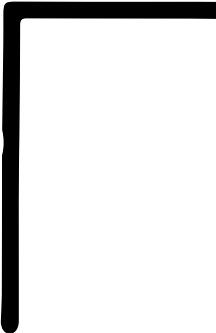
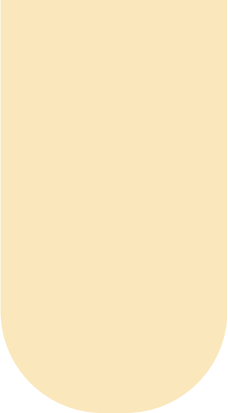
pic.5. Checkmate. Game over



pic.6. Example of stalemate



pic.7. pawn promotion
example



CONCLUSION

FUTHER DEVELOPMENT

- Implementation of Pattern Command
- points counter
- color selection
- code refactoring)
- menu



THANK YOU